Physical change

- doesn't produce any new chemical substances
- often easy to reverse

e.g making a mixture or dissolving a solute

Chemical change

- produces new chemical substances
- product has different properties to their reactant.
- very difficult to reverse
- mostly exothermic

e.g color change / bubbles of gas / precipitate

Rates of reaction factors :

→ concentration
more concentration = faster rate of reaction
→ pressure
more pressure = faster rate of reaction
→ temperature
higher temp. = faster rate of reaction
→ surface area (of solid reactants)
more surface area = faster rate of reaction
→ catalyst (including enzyme)
presence of catalyst = faster rate of reaction

* a catalyst speeds up the rate of a reaction without being consumed by it
* a catalyst decreases the activation energy, E_a, of a reaction

Collision theory

- for a reaction to occur, particles must collide
- not only that, the collision must have sufficient energy to break bonds

→ minimum energy that colliding particles need is known as activation energy (E_a)

Investigating the rate of reaction:

- we need to measure either how quickly the reactants are used up or how quickly the products are formed.

→ practical methods
- measure change in mass
- measure volume of gas produced
- measure with/without catalyst

Graphs

- steeper the curve, faster the rate of reaction
- rate is quickest at the beginning of the reaction
- as reaction progresses, concentration of reactants decreases thus rate decreases
- when one of the reactants is used up, the reaction stops.

Reversible reactions

- some chemical reactions are reversible
- the symbol used is "$\rightleftharpoons$"

conditions :-

(1) effect of heat on hydrated compounds
(2) addition of water to anhydrous compounds

$$\underbrace{\underset{\text{copper sulfate (blue)}}{CuSO_4} \cdot \underset{\text{\& water}}{5H_2O}}_{\text{hydrated}} \underset{\text{(heat)}}{\rightleftharpoons} \underbrace{\underset{\text{copper sulfate (white)}}{CuSO_4}}_{\text{anhydrous}} + \underset{\text{water}}{5H_2O}$$

exothermic ⟵ ⟶ endothermic

$$\underbrace{\underset{\text{cobalt chloride (pink)}}{CoCl_2} \cdot \underset{\text{\& water}}{6H_2O}}_{\text{hydrated}} \underset{\text{(heat)}}{\rightleftharpoons} \underbrace{\underset{\text{cobalt chloride (blue)}}{CoCl_2}}_{\text{anhydrous}} + \underset{\text{water}}{6H_2O}$$

a reversible reaction in a closed system is at equilibrium when:-

* rate of forward reaction = rate of reverse reaction
* concentrations of reactants & products is no longer changing.

Equilibrium

- dynamic; molecules on the left & right of the equation are changing into each other at the same rate
- concentration of the reactants & products remain same given there is no change in temp. or pressure
- only occurs in a closed system

in a reversible reaction, how is the position of equilibrium affected by:

(1) change in temperature
(2) change in pressure
(3) change in concentration
(4) use of catalyst

- change in temperature

* think of temperature as a part of the reaction

$PCl_3 + Cl_2 \rightleftharpoons PCl_5$ (+ 60 kJ)

→ temp. ↑

↳ position of equilibrium to the left

N_2O_4 (+ 57 kJ) $\rightleftharpoons 2NO_2$

→ temp. ↓

↳ position of equilibrium to the left

- change in pressure

* when pressure is increased, the equilibrium shifts to the side with the fewer number of moles of gas

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

→ pressure ↑

↳ position of equilibrium to the right

- change in concentration

$H_2 + N_2 \rightleftharpoons 2NH_3$

→ concentration ↑ (N_2)

↳ position of equilibrium to the right

- use of catalyst

* doesn't affect position of equilibrium, helps reach equilibrium faster

Haber process

- used for the manufacture of Ammonia

→ $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
nitrogen + hydrogen ammonia

$\boxed{\text{conditions:}}$

- 450°C
- 20 000 kPa / 200 atm
- iron catalyst

* the nitrogen is obtained by seperating it from the air
* the hydrogen is obtained from natural gases such as methane

Contact process

- used to make sulfuric acid

→ $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
sulfur dioxide + oxygen sulfur trioxide

turning sulfur dioxide ~~to trioxide for the~~ contact process

$\boxed{\text{conditions:}}$

- 450°C
- 200 kPa / 2 atm
- vanadium (V) oxide catalyst

Explaining these conditions

- Haber process

* 450°C temperature because it moves the equilibrium to the right, making more ammonia but since it is a Low temperature, the rate of reaction is slower, So a compromise is made.

* **20 000 kPa / 200 atm** is used but it is dangerous and it is very expensive but more product is made since it's indeed a lot of pressure.
* **iron catalyst** is used because its cheap, durable and lowers "E_a" so the process can occur at lower temperatures.

- Contact process
* **450°C** temperature is used because the reaction is exothermic so that a low temperature would produce more sulfur trioxide (SO_3)
* **200 kPa / 2 atm** is used because the equilibrium position is already far to the right so there is no need for high pressure, not too expensive either
* **vanadium (V) oxide catalyst** is used since it increases the rate of reaction thus reaching the equilibrium position faster.

Redox reaction

- a reaction in which oxidation and reduction happen at the same time.
- oxidation is a reaction in which oxygen is added.
- reduction is a reaction in which oxygen is removed.
- involve loss/gain of oxygen and electrons

$$\overbrace{2Al_2O_3 + \underbrace{3C \rightarrow 4Al}_{}}^{\text{reduction}}\underbrace{+\ CO_2}_{\text{oxidation}}$$

Roman numerals for oxidation number :-

ZnO → zinc oxide

CrO → chromium (II) oxide

Cr_2O_3 → chromium (III) oxide

CrO_3 → chromium (VI) oxide

* oxidation is the loss of electrons and increase of oxidation number
* reduction is the gain of electrons and decrease of oxidation number

Changes in oxidation number :-

→ oxidation number of elements in their uncombined state is 0

→ oxidation number of a monatomic ion is the same as the charge on the ion

→ sum of oxidation numbers in a compound is 0

→ sum of oxidation numbers in an ion is equal to the charge on the ion.

Color changes :-

- so potassium manganate (VII) is an oxidising agent that when added to reducing agent, it's purple color changes to colorless.
- potassium iodide (aq) is a reducing agent that when added to an oxidising agent, turns the solution red-brown.

→ an oxidising agent oxidises another substance and itself is reduced

→ a reducing agent reduces another substance and itself is oxidised

Identifying oxidising/reducing agent:

① • $Fe + Br_2 \rightarrow FeBr_2$

• half equations

↳ $Fe \rightarrow Fe^{2+} + 2e^-$ (Fe loses electron)

↳ $Br_2 + 2e^- \rightarrow 2Br^-$ (Br gains electron)

• Fe = oxidised

• Br = reduced

• reducing agent = Fe since it oxidised itself and caused Br_2 to get reduced

② • $Cl_2 + 2KI \rightarrow 2KCl + I_2$

• half equations

↳ $Cl_2 + 2e^- \rightarrow 2Cl^-$ (Cl gains electron)

↳ $2K^+ + 2e^- \rightarrow 2K$ (K gains electron)

↳ $2I^- \rightarrow I_2 + 2e^-$ (I loses electron)

• Cl = reduced

• K = N/A

• I = oxidised

• oxidising agent = Cl since it reduced itself and caused I to get

O	I	L	R	I	G
oxidation	is	loss	reduction	is	gain

of electrons